VICTOR DANIEL RAMOS BESSA

São José dos Campos, SP - Brazil | (12) 988176182 | victordanielramosb@gmail.com |
github.com/victordanielrb

SUMMARY

Full-stack software engineer and startup co-founder, running a production AI product end to end: NestJS backend, queue-based processing with Redis and BullMQ, PostgreSQL, React frontend, self-managed infrastructure and production incident resolution. Experienced in LLM agent systems (Google ADK), multimodal retrieval and REST API design. Familiar with Agile/Scrum, microservices and CI/CD.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, SQL
- **Backend:** NestJS, Node.js (Express), Flask, Spring Boot, REST APIs, microservices, JWT
- **Frontend:** React, React Native (Expo), Next.js, Tailwind CSS
- **Data & Messaging:** PostgreSQL, MongoDB, Redis, BullMQ, sqlite-vec (vector search)
- **AI/ML:** LLM agents (Google ADK), multimodal embeddings (Gemini), cosine-similarity retrieval, LLM reranking, TensorFlow, Essentia
- **DevOps:** Coolify, Hetzner Cloud, GitHub Actions (CI/CD), production deployments and incident response
- **Other:** Agile/Scrum, adaptability, proactivity, analytical thinking. **Spoken:** English (C2), Spanish (A2)

EXPERIENCE

Bubble Labs – Co-Founder & Full-Stack Developer **May 2026 - Present**

bubblelabs.app – AI startup building **Prisma**, a conversational sales and customer-service agent for WhatsApp and Instagram that qualifies leads, schedules appointments and closes sales with integrated Pix payments. Currently serving one client and expanding.

- Develop the backend in NestJS with PostgreSQL, taking tasks from specification through implementation to production.
- Built asynchronous processing on Redis and BullMQ, making heavy use of job queues to handle background workloads reliably.
- Developed conversational AI agents with Google ADK for lead qualification, scheduling and sales flows.
- Built the React frontend and integrated it with the backend APIs.
- Own DevOps: deploy and operate the platform on Hetzner servers with Coolify, handling frequent releases and hotfixes.
- Diagnose and resolve production incidents to keep the client's customer-facing service available.
- Translate client requirements into technical tasks and contribute to product decisions as a co-founder.

PROJECTS

VIBEZ **Jun 2026 - Jul 2026**

Multimodal music recommendation system that maps the visual semantics of an image to the audio characteristics of YouTube playlist tracks, using Gemini multimodal embeddings and cosine-similarity search.

- Architected the system from scratch, benchmarking four ML and embedding approaches before selecting the final design.
- Evolved retrieval across three iterations: direct cross-modal embedding, a textual bridge, and two-stage retrieval with LLM reranking.
- Designed the AI pipeline: image description, genre extraction, dual vector embeddings, cosine-similarity retrieval and multimodal LLM reranking.
- Implemented modular Google ADK agents (`image_describer`, `genre_extractor`, `track_reranker`).
- Integrated Gemini embeddings, Essentia DSP, EffNet-Discogs (TensorFlow), sqlite-vec, yt-dlp and ffmpeg.

EDUCATION

Multiplatform Software Development – FATEC São José dos Campos **April 2024 - (Est.) Jul 2027**